

Supplementary Material

Imaging Informatics for Thermal Ablation: CT Thermometry–Based Tissue Injury Mapping

S1. Cooling-phase record and paired dose sensitivity

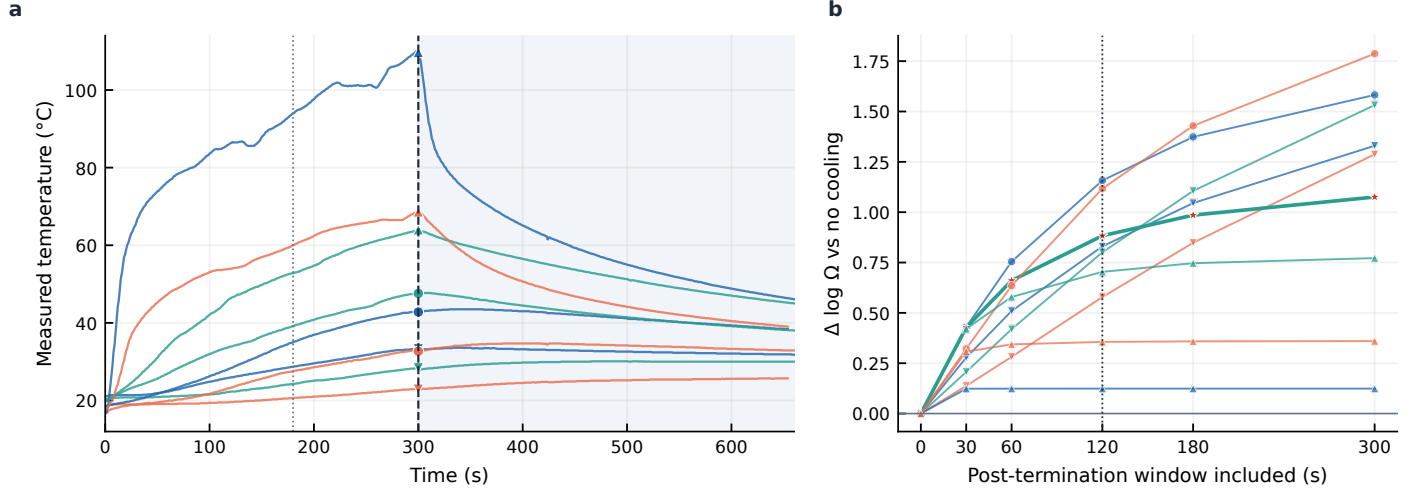


Figure S1: Measured 300-s post-termination cooling sensitivity. (a) Nine ROI trajectories with the 300-s power-termination marker and the available post-marker measured window. Colors encode the three radial distances from the antenna. (b) Additional $\log \Omega$ contributed by progressively longer measured post-termination windows relative to the no-cooling estimate.

Table S1: Cooling-phase record and analysis protocol.

Item	Recorded information and analysis choice
Endpoint cohorts	Nine ROI series from the 300-s (50 W, 5-min) protocol; no 180-s post-termination series was supplied
Time axis	Seconds from the start of the record; one value per second where an ROI value is available
Cooling marker	$t = 300$ s, used as the power-termination marker for this sensitivity analysis

Table S2: Cooling-inclusive versus no-cooling Arrhenius dose estimates. Values are mean $\pm$ SD across available ROIs; $\Delta \log \Omega = \log \Omega_{\text{cool}} - \log \Omega_{\text{no cool}}$.

Endpoint marker	n	Available post-marker record	$\Delta \log \Omega$, 30 s	$\Delta \log \Omega$, 60 s	$\Delta \log \Omega$, 120 s	$\Delta \log \Omega$, 180 s	$\Delta \log \Omega$, 300 s	Crossing at 120 s
180 s	0	Not recorded	–	–	–	–	–	Not assessable
300 s	9	354–832 s	0.295 ± 0.120	0.478 ± 0.203	0.728 ± 0.336	0.891 ± 0.432	1.094 ± 0.569	1/9 (ROI 5)

Table S3: Per-ROI log-dose values for the 300-s sensitivity analysis.

ROI	Distance (mm)	$\log \Omega_{\text{no cool}}$	$t_c = 30$ s	$t_c = 60$ s	$t_c = 120$ s	$t_c = 180$ s	$t_c = 300$ s	Crossing
1	3.5	-4.7953	-4.5173	-4.2859	-3.9653	-3.7501	-3.4653	–
2	9.0	-6.1213	-5.9145	-5.7023	-5.3207	-5.0157	-4.5897	–
3	14.5	-7.4923	-7.3548	-7.2107	-6.9138	-6.6444	-6.2051	–
4	3.5	-2.1561	-1.7259	-1.4014	-0.9988	-0.7822	-0.5738	–
5	9.0	-0.8324	-0.4047	-0.1721	0.0506	0.1529	0.2430	120 s
6	14.5	-4.9604	-4.6390	-4.3255	-3.8435	-3.5314	-3.1733	–
7	3.5	14.0546	14.1780	14.1784	14.1785	14.1785	14.1785	Already > 0
8	9.0	3.5253	3.9457	4.1043	4.2291	4.2720	4.2972	Already > 0
9	14.5	4.6934	5.0011	5.0369	5.0492	5.0518	5.0532	Already > 0

S2. Experimental hardware and data flow

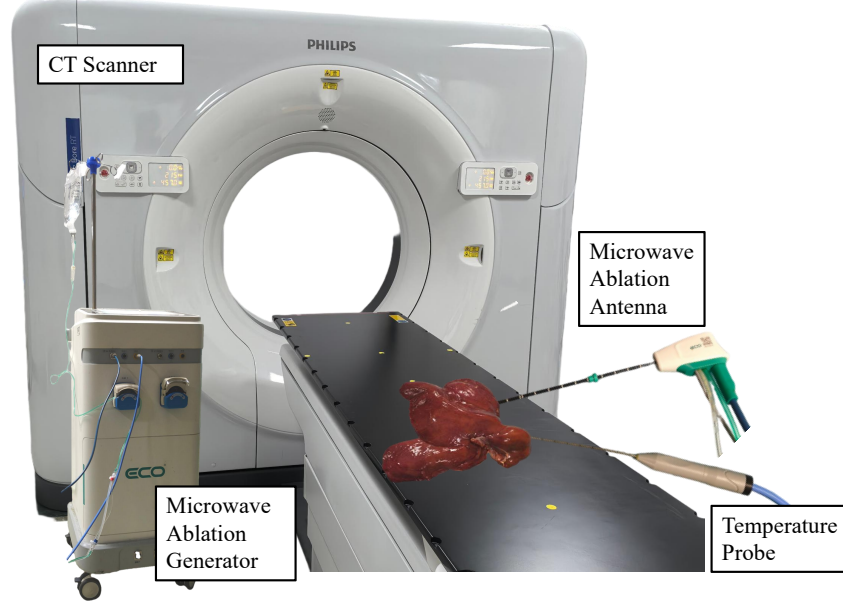


Figure S2: Experimental hardware configuration for CT-guided microwave ablation.

Table S4: Study-level allocation of specimens, acquisition endpoints, and model roles. The 180-s and 300-s endpoints were generated from separate 50-W 3-min and 5-min experiments. An additional five-liver porcine cohort underwent the 300-s protocol followed by post-termination cooling measurements.

Cohort	Number	Endpoint/protocol	Used for
Porcine CT-thermometry development	15 blocks	50 W; calibration	HU-temperature calibration
Porcine injury-model development	15 blocks	50 W; 180/300 s	27 ROI-level curves, temperature templates, and Arrhenius mappings
Porcine 300-s cooling-experiment cohort	5 livers	50 W; 300-s heating followed by post-termination cooling	Measured post-termination cooling trajectories, cooling-inclusive Arrhenius dose estimation, and cooling-related boundary-sensitivity analysis
Porcine final validation	10 blocks	Independent 180/300 s endpoints	Final thermometry and injury-map evaluation
Bovine exploratory evaluation	2 cases	3/5-min endpoints	Cross-tissue exploratory analysis

Table S5: Auditable specimen, batch, and curve identifiers available in the supplied data export. Block-level IDs were not preserved in the export; therefore, no block identity is inferred from the ROI-level files.

Analysis role	Blocks	IDs in supplied export	Endpoint assignment	Curve/ROI mapping
CT thermometry development	15	Case/Record ID available; block ID absent	Calibration records	ROI-level HU-T pairs
Temperature-history development	15	Block ID absent	Independent 180/300-s endpoints	Exp1/2/3, each 9 ROIs
Independent porcine validation	10	Validation case ID available; block ID absent	Independent 180/300-s endpoints	Final gross-boundary evaluation
Bovine exploratory cases	2	Bovine case ID available	3-min and 5-min endpoints	Cross-tissue exploratory evaluation

S3. CT thermometry calibration audit

$$T = 99.3013 - 0.8416 \text{ HU}, \quad \Delta T = -0.8416 \Delta \text{HU}. \quad (1)$$

Table S6: Calibration sample accounting and performance sensitivity.

Dataset	n	Model/status	R^2	RMSE (C)	MAE (C)
Merged records	117	Record accounting	–	–	–
Complete pairs	114	Complete HU–T pairs	–	–	–
All eligible pairs	58	$T = 77.8677 - 0.5525 \text{ HU}$	0.4407	8.6544	7.0525
Retained domain	39	$T = 99.3013 - 0.8416 \text{ HU}$	0.8749	3.8837	3.1661
Screened observations	19	Externally studentized residual criterion	–	–	–

Table S7: Distribution of the 58 eligible calibration observations before and after screening.

Grouping	Eligible	Retained	Screened	Screened fraction
Case 1	14	13	1	7.1%
Case 2	9	4	5	55.6%
Case 3	15	6	9	60.0%
Case 4	20	16	4	20.0%
Time 0 min	16	16	0	0.0%
Time 3 min	27	15	12	44.4%
Time 5 min	15	8	7	46.7%

Table S8: All 58 eligible calibration observations within 20–60 °C.

Record	Case	Time	HU	T (C)	Status
CT0009	1	3	25.00	58.20	Screened
CT0029	2	3	61.79	58.37	Screened
CT0032	2	3	66.70	56.13	Screened
CT0036	2	3	53.00	40.63	Screened
CT0041	2	5	57.57	59.83	Screened
CT0045	2	5	50.69	34.43	Screened
CT0057	3	3	68.52	29.90	Screened
CT0060	3	3	63.48	28.20	Screened
CT0063	3	3	53.00	30.30	Screened
CT0065	3	3	72.67	52.10	Screened
CT0068	3	5	45.20	48.80	Screened
CT0069	3	5	57.59	32.50	Screened
CT0072	3	5	54.00	35.60	Screened
CT0074	3	5	72.38	56.70	Screened
CT0081	3	5	50.69	39.90	Screened
CT0091	4	3	83.00	43.20	Screened
CT0098	4	3	71.00	56.40	Screened
CT0100	4	3	73.00	48.20	Screened
CT0102	4	3	76.00	26.80	Screened
CT0019	1	0	76.00	37.00	Retained
CT0020	1	0	82.00	37.00	Retained
CT0021	1	0	78.00	37.00	Retained
CT0022	1	0	77.00	37.00	Retained
CT0023	1	0	75.00	37.00	Retained
CT0024	1	0	69.00	37.00	Retained
CT0025	1	0	72.00	37.00	Retained
CT0026	1	0	79.00	37.00	Retained
CT0027	1	0	73.00	37.00	Retained
CT0002	1	3	49.00	56.50	Retained
CT0003	1	3	58.00	50.60	Retained
CT0005	1	3	44.00	59.10	Retained
CT0006	1	3	60.00	55.20	Retained

Record	Case	Time	HU	T (C)	Status
CT0030	2	3	68.52	39.25	Retained
CT0033	2	3	63.48	38.89	Retained
CT0039	2	5	81.70	35.04	Retained
CT0042	2	5	72.10	36.31	Retained
CT0056	3	3	61.79	46.80	Retained
CT0059	3	3	66.70	44.30	Retained
CT0066	3	3	82.13	34.60	Retained
CT0075	3	5	81.70	37.90	Retained
CT0077	3	5	57.57	51.80	Retained
CT0078	3	5	72.10	35.50	Retained
CT0109	4	0	91.00	21.50	Retained
CT0111	4	0	85.00	20.40	Retained
CT0112	4	0	93.00	20.20	Retained
CT0113	4	0	89.00	21.40	Retained
CT0114	4	0	85.00	20.70	Retained
CT0115	4	0	94.00	21.10	Retained
CT0117	4	0	86.00	20.70	Retained
CT0092	4	3	87.00	26.90	Retained
CT0093	4	3	84.00	23.60	Retained
CT0095	4	3	67.00	49.70	Retained
CT0096	4	3	81.00	34.10	Retained
CT0099	4	3	78.00	37.00	Retained
CT0101	4	3	77.00	31.00	Retained
CT0104	4	5	49.00	57.40	Retained
CT0105	4	5	68.00	43.00	Retained
CT0108	4	5	61.00	49.90	Retained

Table S9: Details of the 19 screened calibration observations.

Record	Case	Time	HU	T (C)	Iter.	r_{ext}	Threshold
CT0063	3	3	53.00	30.30	1	-2.191	2.003
CT0074	3	5	72.38	56.70	1	2.232	2.003
CT0098	4	3	71.00	56.40	1	2.095	2.003
CT0032	2	3	66.70	56.13	2	2.013	2.006
CT0045	2	5	50.69	34.43	2	-2.109	2.006
CT0029	2	3	61.79	58.37	3	2.030	2.008
CT0060	3	3	63.48	28.20	3	-2.050	2.008
CT0065	3	3	72.67	52.10	3	2.081	2.008
CT0041	2	5	57.57	59.83	4	2.138	2.011
CT0069	3	5	57.59	32.50	4	-2.140	2.011
CT0072	3	5	54.00	35.60	5	-2.170	2.013
CT0091	4	3	83.00	43.20	5	2.122	2.013
CT0081	3	5	50.69	39.90	6	-2.089	2.015
CT0100	4	3	73.00	48.20	6	2.095	2.015
CT0009	1	3	25.00	58.20	7	-2.512	2.018
CT0036	2	3	53.00	40.63	8	-2.453	2.020
CT0057	3	3	68.52	29.90	8	-2.155	2.020
CT0068	3	5	45.20	48.80	9	-2.745	2.023
CT0102	4	3	76.00	26.80	10	-2.114	2.024

Table S10: Leakage-controlled calibration validation. Screening, when used, was recomputed from the training fold only; the held-out observations remained eligible test data.

Analysis	n	R^2	RMSE (C)	MAE (C)
Leave-one-out, unfiltered	58	0.4027	8.9436	7.3090
Leave-one-out, training-fold screening	58	0.2927	9.7323	7.3051
Leave-one-case-out, unfiltered	58	0.2565	9.9781	8.2152
Leave-one-case-out, training-fold screening	58	0.1854	10.4443	8.6103

S4. Temperature-curve quality control and templates

Table S11: Hard-QC outcome and experiment-wise Tukey diagnostics. The listed exclusions were hard-QC exclusions; no additional curve was excluded by the Tukey diagnostic.

Model	Experiment	Before	Hard-QC excluded	Retained	Tukey outliers	Reason
180 s	Exp1	9	1	8	0	Full-interval rollback
180 s	Exp2	9	1	8	0	Endpoint rise = 1.6 °C
180 s	Exp3	9	0	9	0	–
300 s	Exp1	9	1	8	0	Full-interval rollback
300 s	Exp2	9	0	9	0	–
300 s	Exp3	9	0	9	0	–

Table S12: Key nodes of the normalized median heating templates and uncertainty bands for the 180- and 300-s models.

t (s)	$s_{\text{med},180}$	$P_{25}-P_{75}$	$P_{10}-P_{90}$	$s_{\text{med},300}$	$P_{25}-P_{75}$	$P_{10}-P_{90}$
0	0.0000	0.0000–0.0000	0.0000–0.0000	0.0000	0.0000–0.0000	0.0000–0.0000
30	0.1869	0.1166–0.2742	0.1035–0.5160	0.1299	0.0863–0.1929	0.0662–0.4838
60	0.3619	0.2900–0.5410	0.2527–0.7757	0.2521	0.1926–0.4585	0.1535–0.6998
90	0.5484	0.4800–0.7623	0.4290–0.8782	0.3951	0.3122–0.6236	0.2437–0.7678
120	0.7178	0.6813–0.8701	0.6170–0.9318	0.5454	0.4266–0.7193	0.3670–0.8063
150	0.8774	0.8450–0.9425	0.8193–0.9731	0.6581	0.5328–0.7914	0.4843–0.8410
180	1.0000	1.0000–1.0000	1.0000–1.0000	0.7417	0.6400–0.8376	0.5808–0.8811
210	–	–	–	0.8077	0.7497–0.8806	0.6848–0.9238
240	–	–	–	0.8704	0.8452–0.9308	0.7938–0.9586
270	–	–	–	0.9376	0.9256–0.9768	0.9057–0.9838
300	–	–	–	1.0000	1.0000–1.0000	1.0000–1.0000

S5. Runtime profiling

Table S13: Per-run and per-module runtime profiling of the visualization pipeline (ms).

Run	Total	HU load	HU→T	Upd. T	T→logΩ	Upd. logΩ	Overlay	3D VR
1	1688.0	32.5	113.2	28.9	1256.0	60.3	54.9	142.1
2	1652.4	26.6	96.8	32.2	1281.4	58.9	85.4	71.1
3	923.6	14.4	47.3	15.0	652.2	42.5	144.9	7.1
4	935.0	20.0	52.9	14.8	660.0	42.9	38.5	105.9
5	793.7	11.2	39.8	11.7	535.3	33.7	40.4	121.7
Mean±SD	1198.5±434.3	20.9±8.7	70.0±32.8	20.5±9.3	877.0±361.1	47.7±11.5	72.8±44.5	89.6±52.9

S6. Case-wise boundary validation

Table S14: Case-wise comparison of the CT-derived 60 °C isotherm and the proposed injury-map delineation against the matched gross-section transition-zone reference. $|\Delta L|$ and $|\Delta W|$ denote absolute major- and minor-axis errors, respectively. Dice and HD95 quantify post-alignment two-dimensional contour agreement in the matched plane rather than absolute localization accuracy.

Tissue	Case	Time (min)	60 °C isotherm				Injury map			
			$ \Delta L $ (mm)	$ \Delta W $ (mm)	Dice	HD95 (mm)	$ \Delta L $ (mm)	$ \Delta W $ (mm)	Dice	HD95 (mm)
Porcine	1	3	1.75	2.29	0.9274	2.1709	1.10	1.32	0.9125	2.0714
Porcine	2	3	1.61	0.09	0.8835	2.7001	2.89	3.29	0.9250	1.8282
Porcine	3	3	2.03	0.16	0.9085	2.9003	1.06	0.62	0.9230	1.8548
Porcine	4	3	1.62	0.16	0.9380	2.2048	0.97	3.18	0.9345	2.2779
Porcine	5	3	2.33	1.35	0.9078	2.7691	0.71	1.20	0.9314	1.8467
Porcine	6	5	3.80	2.56	0.9489	1.9754	0.54	4.24	0.9393	2.1200
Porcine	7	5	1.19	0.70	0.9123	2.8394	0.28	0.86	0.9118	2.1223
Porcine	8	5	0.51	0.34	0.8829	2.8154	0.42	0.27	0.8858	2.5746
Porcine	9	5	2.52	3.89	0.9186	2.7339	0.11	2.34	0.9429	1.7458
Porcine	10	5	4.07	3.68	0.9502	1.6758	0.72	0.69	0.9524	1.7480
Bovine	1	3	1.41	1.21	0.7626	3.4477	1.44	1.84	0.8478	2.3484
Bovine	2	5	3.79	0.87	0.8355	3.7501	1.28	3.85	0.8745	2.3272